

Formative Assessment 3: Linear Models (APM1205)

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Preliminary Definitions & Properties

Before starting the proofs, we define the weights m_i for the slope estimator B :

$$m_i = \frac{x_i - \bar{x}}{\sum_{j=1}^n (x_j - \bar{x})^2}$$

These weights exhibit three fundamental properties that will be utilized across the derivations:

1. $\sum_{i=1}^n m_i = \frac{\sum (x_i - \bar{x})}{\sum (x_j - \bar{x})^2} = \frac{0}{\sum (x_j - \bar{x})^2} = 0$
 2. $\sum_{i=1}^n m_i x_i = \frac{\sum (x_i - \bar{x}) x_i}{\sum (x_j - \bar{x})^2} = \frac{\sum (x_i - \bar{x})^2}{\sum (x_j - \bar{x})^2} = 1$
 3. $\sum_{i=1}^n m_i^2 = \sum \left(\frac{x_i - \bar{x}}{\sum (x_j - \bar{x})^2} \right)^2 = \frac{\sum (x_i - \bar{x})^2}{[\sum (x_j - \bar{x})^2]^2} = \frac{1}{\sum_{i=1}^n (x_i - \bar{x})^2}$
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Question 1

Demonstrate the unbiasedness of the least-squares estimators A and B of α and β in simple regression.

(a) **Show that $E(B) = \beta$:**

Given the linear expression $B = \sum_{i=1}^n m_i Y_i$ and the assumption of linearity $E(Y_i) = \alpha + \beta x_i$, we apply the linear property of expectation:

$$\begin{aligned} E(B) &= E\left(\sum_{i=1}^n m_i Y_i\right) \\ &= \sum_{i=1}^n m_i E(Y_i) \quad (\text{by linearity of expectation}) \\ &= \sum_{i=1}^n m_i (\alpha + \beta x_i) \quad (\text{substituting } E(Y_i) = \alpha + \beta x_i) \\ &= \alpha \sum_{i=1}^n m_i + \beta \sum_{i=1}^n m_i x_i \end{aligned}$$

Applying the properties $\sum m_i = 0$ and $\sum m_i x_i = 1$:

$$\begin{aligned} E(B) &= \alpha(0) + \beta(1) \\ &= \beta \end{aligned}$$

Thus, B is an unbiased estimator of β .

(b) **Show that A can be written as a linear function of Y_i , and show that $E(A) = \alpha$:**

The least-squares intercept is defined as $A = \bar{Y} - B\bar{x}$. Expanding this:

$$\begin{aligned} A &= \frac{1}{n} \sum_{i=1}^n Y_i - \left(\sum_{i=1}^n m_i Y_i \right) \bar{x} \\ &= \sum_{i=1}^n \left(\frac{1}{n} - \bar{x} m_i \right) Y_i \end{aligned}$$

Let $k_i = \frac{1}{n} - \bar{x} m_i$. Thus, $A = \sum_{i=1}^n k_i Y_i$, meaning A is a linear function of Y_i .

Now, taking the expectation of A :

$$\begin{aligned} E(A) &= \sum_{i=1}^n k_i E(Y_i) \\ &= \sum_{i=1}^n \left(\frac{1}{n} - \bar{x} m_i \right) (\alpha + \beta x_i) \\ &= \alpha \sum_{i=1}^n \left(\frac{1}{n} - \bar{x} m_i \right) + \beta \sum_{i=1}^n \left(\frac{1}{n} - \bar{x} m_i \right) x_i \end{aligned}$$

We evaluate the two summation terms independently:

- $\sum_{i=1}^n \left(\frac{1}{n} - \bar{x} m_i \right) = \sum_{i=1}^n \frac{1}{n} - \bar{x} \sum_{i=1}^n m_i = 1 - \bar{x}(0) = 1$
- $\sum_{i=1}^n \left(\frac{1}{n} - \bar{x} m_i \right) x_i = \frac{1}{n} \sum_{i=1}^n x_i - \bar{x} \sum_{i=1}^n m_i x_i = \bar{x} - \bar{x}(1) = 0$

Substituting these results back into our expectation equation yields:

$$E(A) = \alpha(1) + \beta(0) = \alpha$$

Thus, A is an unbiased estimator of α .

Question 2

Derive the sampling variances of A and B in simple regression.

Assumptions used:

- **Constant Variance (Homoscedasticity):** $V(Y_i) = \sigma^2$ for all i .
- **Independence:** $Cov(Y_i, Y_j) = 0$ for all $i \neq j$.

Sampling Variance of B : Because the observations Y_i are independent, the variance of their linear combination contains no covariance terms:

$$\begin{aligned} V(B) &= V \left(\sum_{i=1}^n m_i Y_i \right) \\ &= \sum_{i=1}^n m_i^2 V(Y_i) \quad (\text{due to the independence assumption}) \\ &= \sigma^2 \sum_{i=1}^n m_i^2 \quad (\text{due to the constant variance assumption}) \end{aligned}$$

Substituting property (3) from our preliminary definitions:

$$V(B) = \frac{\sigma^2}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

Sampling Variance of A : Using the linear weights $k_i = \frac{1}{n} - \bar{x}m_i$ established in 1(b):

$$\begin{aligned}
 V(A) &= V\left(\sum_{i=1}^n k_i Y_i\right) \\
 &= \sum_{i=1}^n k_i^2 V(Y_i) \quad (\text{by independence}) \\
 &= \sigma^2 \sum_{i=1}^n \left(\frac{1}{n} - \bar{x}m_i\right)^2 \quad (\text{by constant variance}) \\
 &= \sigma^2 \sum_{i=1}^n \left(\frac{1}{n^2} - \frac{2\bar{x}m_i}{n} + \bar{x}^2 m_i^2\right) \\
 &= \sigma^2 \left(\sum_{i=1}^n \frac{1}{n^2} - \frac{2\bar{x}}{n} \sum_{i=1}^n m_i + \bar{x}^2 \sum_{i=1}^n m_i^2\right)
 \end{aligned}$$

Applying the known properties ($\sum m_i = 0$ and $\sum m_i^2 = \frac{1}{\sum (x_i - \bar{x})^2}$):

$$\begin{aligned}
 V(A) &= \sigma^2 \left(\frac{n}{n^2} - 0 + \frac{\bar{x}^2}{\sum (x_i - \bar{x})^2}\right) \\
 &= \sigma^2 \left[\frac{1}{n} + \frac{\bar{x}^2}{\sum_{i=1}^n (x_i - \bar{x})^2}\right]
 \end{aligned}$$

Question 3

Linear transformation of X and Y in simple regression.

(a) **Transformation $X' = 10(X - 1)$:**

Scaling the independent variable by a factor of $c = 10$ rescales the sample variance of X by c^2 . Since $V(B) = \frac{\sigma^2}{\sum (x_i - \bar{x})^2}$, multiplying the horizontal spread by 10 increases the denominator by a factor of $10^2 = 100$.

$$V(B') = \frac{V(B)}{10^2} \implies SE(B') = \frac{SE(B)}{10}$$

The new slope estimator scales inversely as well ($B' = \frac{B}{10}$). Calculating the t'_0 statistic:

$$t'_0 = \frac{B'}{SE(B')} = \frac{B/10}{SE(B)/10} = \frac{B}{SE(B)} = t_0$$

(b) **Transformation $Y'' = 5(Y + 2)$:**

Scaling the dependent variable by a factor of $d = 5$ directly multiplies the overall model variance σ^2 by $d^2 = 25$.

$$V(B'') = 5^2 \cdot V(B) \implies SE(B'') = 5 \cdot SE(B)$$

The new slope estimator matches this scale shift perfectly ($B'' = 5B$). Calculating the t''_0 statistic:

$$t''_0 = \frac{B''}{SE(B'')} = \frac{5B}{5 \cdot SE(B)} = \frac{B}{SE(B)} = t_0$$

(c) **General effect of linear transformations on inference:**

- **Hypothesis Tests: Completely Unaffected.** Because the t -statistic is a scale-invariant ratio, any scaling applied to the variables cancels out ($t_0 = t'_0 = t''_0$). Consequently, p -values and decisions to reject/fail to reject $H_0 : \beta = 0$ remain identical.
- **Confidence Intervals: Rescaled systematically.** The confidence intervals shift dynamically by the exact scaling factor of the slope (divided by c for X -transformations, or multiplied by d for Y -transformations). While the physical meaning and precision remain intact, the numerical boundaries change to properly match the new units.